

AI-Based Early Health Risk Detection Using Fitbit Wearable Data

Abstract

Wearable devices have emerged as powerful tools for continuous health monitoring by collecting physiological and behavioral data in real time. This project proposes an AI-driven health risk detection system utilizing data collected from Fitbit wearable devices to identify potential health abnormalities before clinical symptoms become apparent.

The system will collect and analyze various biometric indicators including heart rate, heart rate variability, sleep quality, physical activity levels, calorie expenditure, stress indicators, and oxygen saturation measurements. Machine Learning and Deep Learning models such as Random Forest, XGBoost, Convolutional Neural Networks (CNN), and Long Short-Term Memory (LSTM) networks will be developed to detect patterns associated with cardiovascular disorders, stress-related conditions, sleep disorders, and metabolic health risks.

To enhance predictive capability, the project may integrate NLP techniques for analyzing user-generated health logs and symptom descriptions. Computer Vision models can optionally process facial images or posture data captured via smartphone cameras to identify visual indicators of fatigue and stress. The system aims to provide personalized risk assessments, real-time alerts, and health recommendations while maintaining user privacy and data security.

This research contributes to preventive healthcare by leveraging wearable sensor data and advanced AI techniques for continuous health monitoring and early risk detection.

Technologies

- ML: Random Forest, XGBoost
- NN: CNN, LSTM, GRU, Transformer
- NLP: Symptom journal analysis
- CV: Fatigue detection, stress estimation, posture analysis

Datasets

- Fitbit API Data
- WESAD Dataset
- MHEALTH Dataset
- PAMAP2 Dataset
- Sleep Heart Health Study